Customer Segmentation Using RFM Analysis & K-Means Clustering

Understanding Customer Behavior for Strategic Engagement

Objective

- Segment retail customers based on purchasing behavior
- Enable personalized marketing strategies
- Identify high-value customers and churn risks

Project Overview

This project builds a comprehensive **customer segmentation system** that groups retail customers based on purchasing behavior.

Key Benefits

- Understand customer behavior
- Identify valuable customer groups
- Improve marketing targeting
- Increase customer lifetime value

The system combines **RFM analysis, rule-based segmentation, and machine learning clustering** to create actionable customer segments.

Customer Segmentation Workflow

The project follows a structured data science pipeline.

01

Data Collection

Retail transaction dataset with Customer ID, Invoice Date, Quantity, and Unit Price fields

02

Data Cleaning

Remove invalid records, handle missing customer IDs, and calculate total transaction value

03

Feature Engineering

Generate RFM metrics from transaction data to capture customer behavior patterns

RFM Feature Engineering

RFM analysis transforms raw transaction data into behavioral metrics that represent **customer engagement and value**.

Recency (R)

Measures how recently a customer made a purchase

Frequency (F)

Measures how often a customer purchases

Monetary (M)

Measures the total amount spent by the customer

Dual Segmentation Strategy

The project uses **two complementary segmentation techniques** to identify both exceptional and typical customer behaviors.

Rule-Based Segmentation

Statistical methods identify customers with extreme behavior.

These customers are classified as:

- High Spenders
- Power Shoppers
- Elite Customers

Machine Learning Segmentation

Regular customers are grouped using **K-Means clustering**.

This automatically identifies **behavioral patterns** and creates meaningful customer segments.

Standard Customer Segments

Customers within normal behavioral ranges are grouped into four distinct clusters.

1

Premium Customers (Cluster 3)

Customers who spend the most and purchase frequently

2

Regular Customers (Cluster 0)

Customers with consistent but moderate purchasing behavior

3

Occasional Customers (Cluster 2)

Customers who purchase infrequently

4

Lapsed Customers (Cluster 1)

Customers who have not purchased recently

Special Customer Segments

Some customers show **exceptional purchasing behavior**.

High Spender Customers (Cluster A)

Very high spending but low purchase frequency

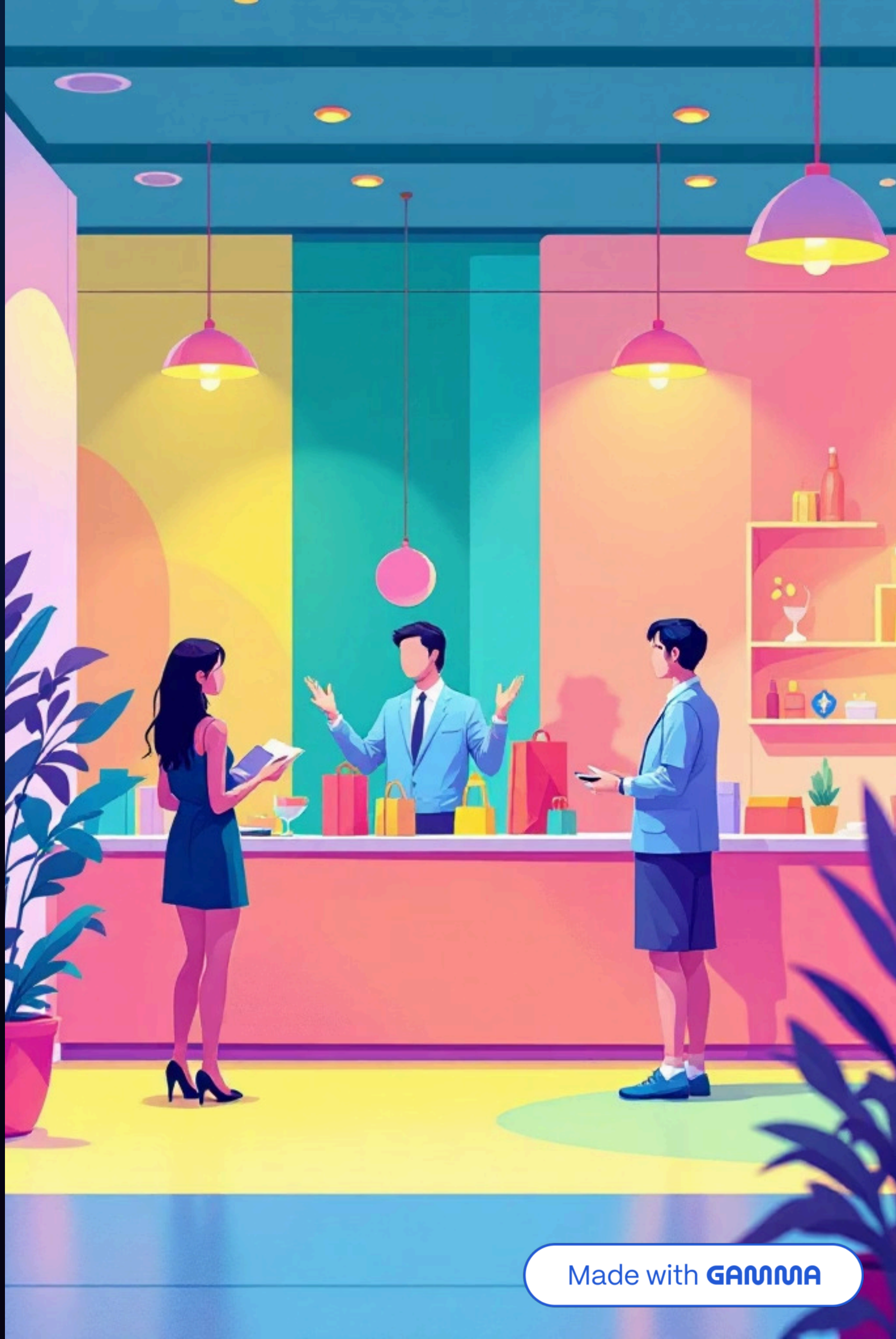
Power Shopper Customers (Cluster B)

Very frequent purchases with moderate spending

Elite Customers (Cluster C)

High spending and high purchase frequency

These segments represent the **most valuable customers**.



Classification Thresholds

Special segments are identified using business thresholds.

High Spender Threshold

Spending greater than

$3,799.39

Power Shopper Threshold

11 or more purchases

Elite Customer

Customers exceeding **both thresholds**

These thresholds help identify **VIP customers before clustering**.

System Architecture & Deployment

The segmentation system is deployed as a **web application**.

Technical Stack

Data Processing

Python, Pandas, NumPy

Machine Learning

Scikit-learn, K-Means, StandardScaler

Visualization

Matplotlib, Seaborn

Web Framework

Flask

Deployment

Render

Users can input customer metrics and instantly receive **segment classification with business insights**.

Business Value & Impact

Customer segmentation enables **data-driven marketing strategies**.

Business Benefits

- Targeted marketing campaigns
- Improved customer retention
- Increased customer lifetime value
- Better customer experience
- Efficient marketing spend